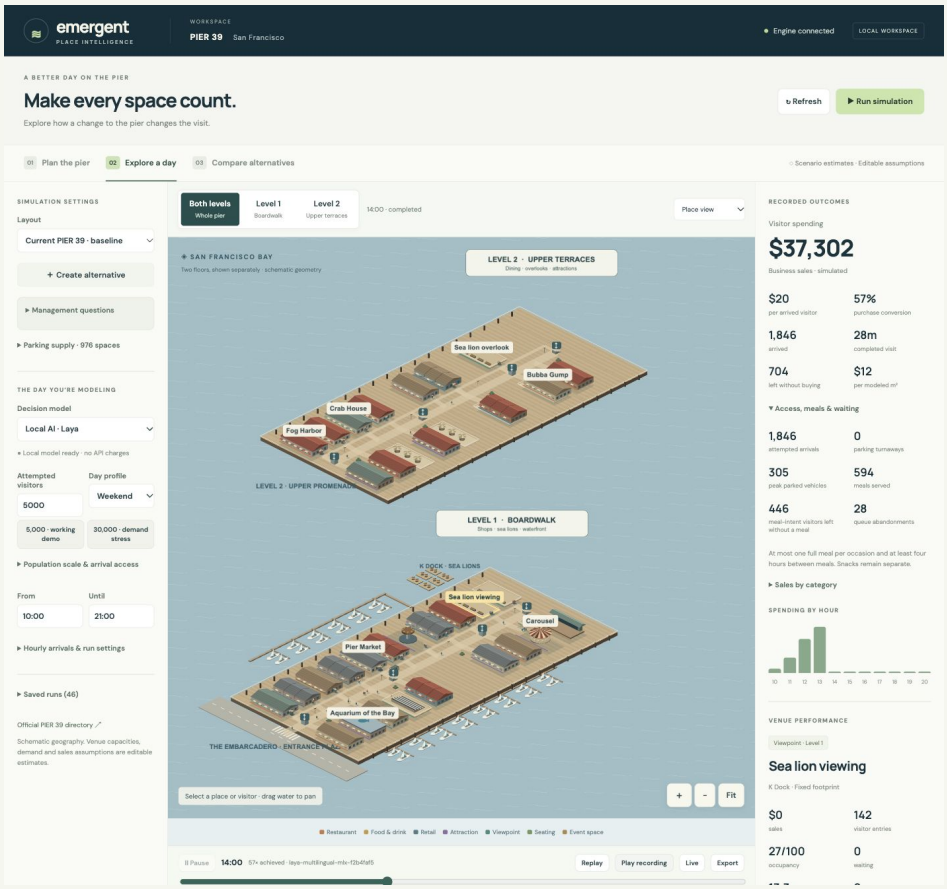
01 / THE PROJECT

Make every space count.

Emergent is a working local planning simulator for PIER 39 management: test the whole visit before changing the place.

THE MANAGEMENT WORKFLOW

Current pier → create alternative → edit → simulate → compare



Actual application: a recorded 5,000-person local-AI scenario at 14:00. Both floors are shown separately, with modeled visitors and recorded outcomes. Graphics and dimensions are illustrative.

The rebuilt experience combines a recognizable waterfront setting with editable tenant uses, operating hours, amenities, pedestrian paths and parking supply. It tracks time, capacity, waiting and purchases in aggregate; restaurant and shop interiors are not simulated.

02 / WHO IT HELPS

One operator. A connected visit.

PIER 39 management can coordinate destination operations, tenant mix and access. The value is seeing tradeoffs across the pier, not just sales at one edited venue.

Decisions with consequences

Management question	Scenario and tradeoff
Add a restaurant?	Convert an upper-level retail cluster into 72 assumed dining places. Test unmet meal demand and sales transferred from other tenants.
Change the tenant mix?	Replace a restaurant with specialty retail, or introduce a cafe near the bay viewpoints. Shopping visits need not produce purchases.
Shift use of space?	Move kiosks, add seating or events, and enable/close or resize internal path segments. Buildings and floor connections remain fixed.
Add or remove parking?	Compare hypothetical supply against the published 976-space garage. Admission changes can expose capacity pressure downstream.
Extend dinner hours?	Keep three restaurants open to 22:00. Both versions receive the same extended arrival window; staffing costs are not included.

Finite meals, finite wallets

Groups have interests, budgets, available time, meal intent and familiarity. A full restaurant purchase satisfies a meal occasion: visitors cannot buy repeated dinners just because another restaurant exists. Another full meal requires at least four hours and enough hunger. Cafes represent snacks or light refreshments. All purchases conserve the group wallet.

Read the pier-wide outcome

The results show total and per-visitor spending, conversion, sales by venue/category/hour, meals, admitted visitors, parking turnaways, dwell, queues and pedestrian activity. Venue gains and declines reconcile to the net sales difference. A new tenant can do well while nearby tenants lose sales; an alternative is allowed to perform worse.

KEEP THE ECONOMICS CLEAR

Visitor spending is business sales. Management income, rent, profit, construction cost and investment returns require separate inputs. This prototype does not infer them.

03 / TECHNOLOGY AND BEHAVIOR

A real engine, with local AI.

The application separates its simulation engine, decision model and interactive interface. It runs locally without paid hosting or an additional paid API.

Layer	What runs
Experience	React, Vite and PixiJS: isometric storefronts, boardwalks, terraces, marina context, two-floor overview, editing, heatmaps and replay.
Engine	TypeScript on Node: 15-second steps, floor-aware routing, venue capacity, queues, dwell, parking, meal limits, purchases and paired experiments.
Intelligence	Local Laya MLX predicts probabilities over feasible destinations. A seeded draw chooses the action. Explicit rules mode is available for faster tests.
Persistence	SpacetimeDB stores layout versions, experiments and compact run state/indexes. Large checkpoints and detailed replays are compressed atomic local files.

Comparable experiments

Cohorts are generated independently of layout. Each pair has the same visitor characteristics, arrivals, mode, budgets and seed. Geometry affects routes, queues and choices. Parking can change admissions while the attempted cohort stays identical. One pair is illustrative; three pairs show seed variation and retain failed/incomplete runs.

What the model considers

The engine filters destinations by opening hours, reachability, capacity to fit the group, affordability and time remaining. It ranks nearby options using preferences, hunger, fatigue, familiarity and observed waits. Local Laya receives current state and candidate options; it is not instructed to optimize revenue. Full venues may cause waits, another choice or departure.

What an explanation means

The inspector records actions and context: destination selection, budget, waits, purchases and departure. Model probabilities are not generated inner thoughts. A provider error is visible and never silently replaced by rules. Jev remains optional, disabled by default and untested with paid credentials.

Runtime: Node 24.21.0, SpacetimeDB 2.10.2, Python 3.12, and a pinned Laya MLX model/bridge. The included local-AI setup targets Apple Silicon macOS. Simulation ticks run in Node; the database provides persistence.

04 / VERIFIED RESULTS

Larger runs. Honest differences.

These are completed synthetic experiments from the real local services. They establish working behavior and accounting, not predictive accuracy for the real pier.

---

20 focused tests passed, plus four real-checkpoint bridge tests, TypeScript and production build checks. Tests cover geography/access, inverse-projected dragging, cohorts, capacity, queues, budgets, opening times, zero-spend browsing, meal limits, restaurant substitution, parking turnover and financial reconciliation. Browser checks cover both floors, editing/history, saved reports, playback and responsive layouts.

Local-AI comparison: three matched 1,000-person pairs

Pair	Current pier	Edited alternative	Sales difference
1	$24,978	$25,455	+$477
2	$21,996	$22,209	+$213
3	$24,614	$24,826	+$212

This alternative moves the kiosk, changes an upstairs shop cluster into a restaurant and extends Pier Market hours within a common 10:00-22:00 window. The identical-layout local-AI control produced zero difference. The deliberately cancelled comparison remained incomplete with no summary delta.

Scale and operational tradeoffs

5,000-person local-AI pair: current-pier sales $113,769; restaurant alternative $112,304; difference -$1,466. This is one illustrative pair, with exact matched cohorts and reconciled ledgers.

Three 30,000-person rules pairs: reducing parking from 976 to 676 spaces admitted 1,883 fewer visitors on average. Mean sales changed -$191, with a range of -$1,875 to +$1,074. Waiting fell by 145,813 person-minutes. Under these capacity assumptions, access changed much more than spending; this is not evidence for reducing real parking.

The bundled actual 5,000-visitor local-AI run completed its modeled day in 11.5 minutes (57x achieved speed). Engine-backed 30,000-person rules runs took 29-37 seconds before final archive completion. Local-AI tests shared the inference service, so timings reflect concurrent validation load. Every visitor was modeled; dollar totals were not multiplied from a smaller sample.

05 / DEMO, SOURCES AND LIMITS

Ready to explore locally.

Get the project at github.com/LSTu5/emergent. Use a saved recording for an immediate presentation, or run a new experiment with explicit assumptions.

Run the demo

1. From the project directory, run **npm run dev** and open **http://127.0.0.1:4320**.
2. Start with Both levels, then focus Level 1 or Level 2.
3. Create a scenario alternative, edit and save it. Dragging validates overlap and access; undo/redo/reset preserve reviewability.
4. Run one or three pairs, or open a saved report. Read venue gains, declines, access and waiting as well as total spending.
5. Inspect A/B, press Replay, then Play recording or scrub the timeline. The inspector shows historical state and actions.

The default is local AI, 5,000 attempted visitors and 10:00-21:00, with venue-specific hours. Demand presets, weekday/weekend weights, car share and parking occupancy are assumptions. A fresh clone first needs **npm ci** and **npm run setup:local**. Preserve both **.local** and **.data** for saved development history.

Source facts versus assumptions

The official 2026 map informs named destinations, floor assignments and broad geography. Published hours inform category defaults. The operator leasing page reports about 11 million visitors in 2023 and 976 garage spaces. All were checked 8 November 2026.

Eleven million divided by 365 is roughly 30,000/day, but this is arithmetic context, not measured demand for an 11-hour window or a season. Dimensions, capacities, prices, conversion, dwell, demand and behavior remain editable estimates. Smaller shops are grouped. Scenery is illustrative; it does not add simulated capacity.

What management would need next

Calibrate with surveyed paths and floors, hourly entrance counts, parking occupancy/turnover, venue capacity and dwell observations, queues and aggregate sales. Validate on unseen days and test multiple plausible inputs. The parking module does not simulate street traffic, overflow garages, fees or construction feasibility. There is no indoor, live transaction, rent, profit or causal uplift model.

Reproduce: npm test; npm run build; npm run test:e2e; npm run test:scale; npm run test:parking. Detailed evidence is in docs/verification*.json, docs/browser-verification.json and data/demo-manifest.json. Three seeds do not establish a forecast confidence interval.